

Quiz 07 v1**Name:** _____*Essential Skills – Must show work for full credit.*

1. Evaluate all six trig functions for $\theta = \frac{5\pi}{3}$.

2. Simplify $\frac{1 - \cos(2\theta)}{\sin^2 \theta}$.

3. Evaluate $\cos\left(\frac{3\pi}{8}\right)$ exactly.

4. Solve $2\sin^2 \theta - 1 = 0$ for $0 \leq \theta < 2\pi$.

Above & Beyond

5. Approximate the area under $f(x) = x^2 + 2$ on $[1, 5]$ using 4 subintervals (left Riemann).

6. Compute a trapezoidal approximation of $f(x) = \sqrt{x + 1}$ on $[2, 6]$ with 4 subintervals.

Quiz 07 v2**Name:** _____*Essential Skills – Must show work for full credit.***1.** Rewrite $\cos(2\theta)$ using sine only.**2.** Evaluate $\sin\left(\frac{\pi}{12}\right)$ exactly.**3.** Solve $\cos \theta = \frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.**4.** Evaluate $\tan\left(-\frac{5\pi}{6}\right)$.

Above & Beyond

5. Approximate the area under $f(x) = x^3 - 1$ on $[0, 4]$ using a right Riemann sum (4 intervals).	6. Explain which is larger for $f(x) = x^2$ on $[1, 5]$: left or right sum.
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Quiz 07 v3**Name:** _____*Essential Skills – Must show work for full credit.*

1. Evaluate $\sin\left(\frac{13\pi}{6}\right)$.

2. Simplify $\sin^2 \theta + \cos^2 \theta$.

3. Evaluate $\cos(105^\circ)$ exactly.

4. Solve $\sec \theta = 2$ on $[0, 2\pi)$.

Above & Beyond

5. Use a trapezoidal sum for $f(x) = \sin x$ on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$.

6. Compare accuracy of left vs trapezoidal approximations.

Quiz 07 v4**Name:** _____*Essential Skills – Must show work for full credit.*

1. Evaluate $\cos\left(-\frac{7\pi}{3}\right)$.

2. Rewrite $\sin(75^\circ)$ using angle sum identities.

3. Solve $\tan \theta = 1$ (general solution).

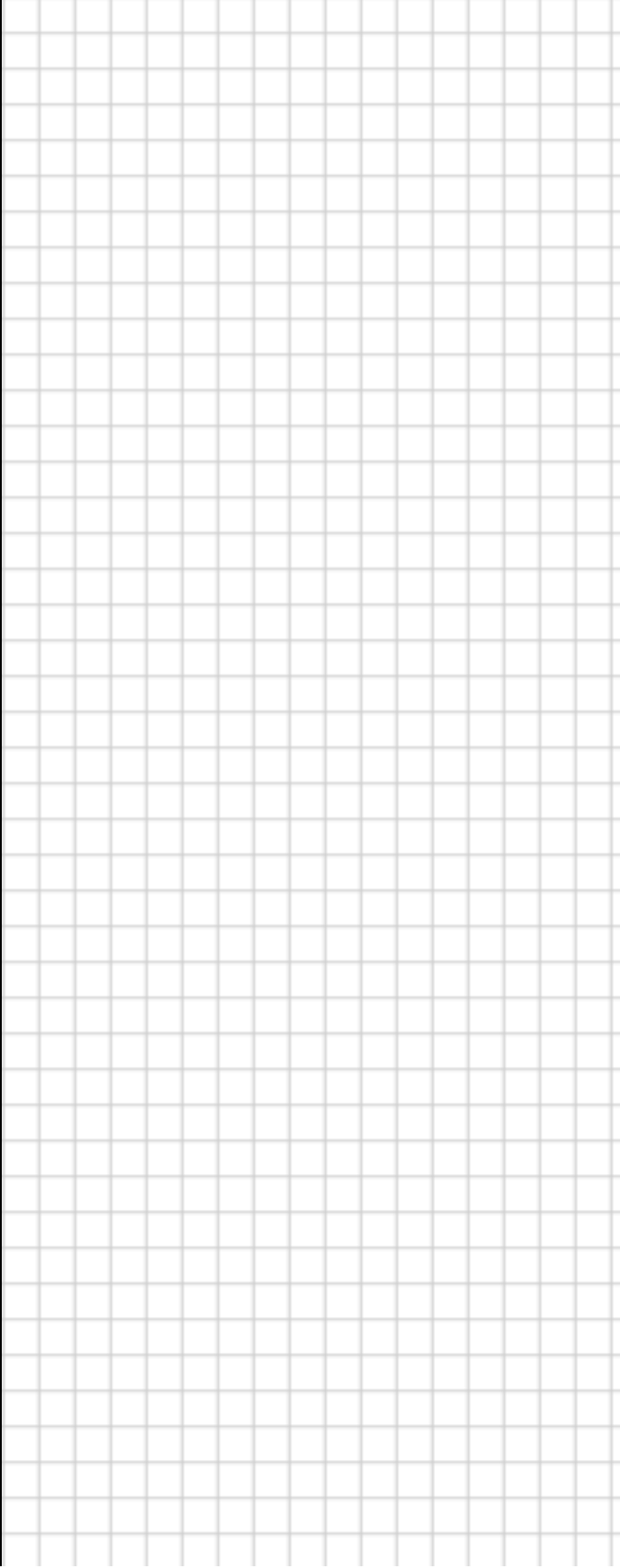
4. If $\sin \theta = \frac{3}{5}$, find $\sin(2\theta)$.

Above & Beyond

5. Approximate area under $f(x) = x^2 - 3x$ on $[1, 5]$ (left sum).



6. Compute trapezoidal estimate for $f(x) = \sqrt{x}$ on $[3, 7]$.



Quiz 07 v5**Name:** _____*Essential Skills – Must show work for full credit.*

1. Evaluate $\tan\left(\frac{11\pi}{4}\right)$.

2. Simplify $\frac{\tan^2 \theta + 1}{\sec^2 \theta}$.

3. Evaluate $\cos\left(\frac{9\pi}{4}\right)$.

4. Solve $2 \cos \theta - 1 = 0$.

Above & Beyond

5. Compute left Riemann sum for $f(x) = \cos x$ on $[0, \frac{\pi}{2}]$.	6. Explain why trapezoidal may be more accurate than left sum.
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Quiz 07 v6**Name:** _____*Essential Skills – Must show work for full credit.*

1. Evaluate $\sin\left(\frac{7\pi}{6}\right)$.

2. Rewrite $\cos(2\theta)$ in terms of cosine only.

3. Evaluate $\tan\left(\frac{5\pi}{4}\right)$.

4. Solve $\cos(2\theta) = 0$ on $[0, 2\pi)$.

Above & Beyond

5. Approximate area under $f(x) = x^2 + x$ on $[2, 6]$ (right sum).

6. Compare left and right sums for $f(x) = x^2$ on $[1, 5]$.

Answer Key — Quiz 07 Versions

Quiz 07 v1

- 1. $\sin = \frac{\sqrt{3}}{2}$, $\cos = \frac{1}{2}$, $\tan = \sqrt{3}$, $\csc = \frac{2\sqrt{3}}{3}$, $\sec = 2$, $\cot = \frac{\sqrt{3}}{3}$
- 2. 2
- 3. $\cos\left(\frac{3\pi}{8}\right) = \frac{\sqrt{2+\sqrt{2}}}{2}$
- 4. $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$
- 5. Left Riemann sum = 60
- 6. Trapezoidal approximation ≈ 8.47

Quiz 07 v2

- 1. $\cos(2\theta) = 1 - 2\sin^2 \theta$
- 2. $\sin\left(\frac{\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$
- 3. $\theta = \frac{\pi}{6}, \frac{11\pi}{6}$
- 4. $\tan\left(-\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{3}$
- 5. Right sum = 84
- 6. Right sum larger if function increasing

Quiz 07 v3

- 1. $\sin\left(\frac{13\pi}{6}\right) = -\frac{1}{2}$
- 2. 1
- 3. $\cos(105^\circ) = -\frac{\sqrt{6}-\sqrt{2}}{4}$
- 4. $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$
- 5. Trapezoidal ≈ 0
- 6. Trapezoidal typically more accurate due to averaging

Quiz 07 v4

- 1. $\cos\left(-\frac{7\pi}{3}\right) = \frac{1}{2}$
- 2. $\sin 75^\circ = \frac{\sqrt{6}+\sqrt{2}}{4}$
- 3. $\theta = \frac{\pi}{4} + k\pi$
- 4. $\sin(2\theta) = \frac{24}{25}$
- 5. Left sum = 24
- 6. Trapezoidal ≈ 7.35

Quiz 07 v5

- 1. $\tan\left(\frac{11\pi}{4}\right) = 1$
- 2. 1
- 3. $\cos\left(\frac{9\pi}{4}\right) = \frac{\sqrt{2}}{2}$
- 4. $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$
- 5. Left sum ≈ 1.11
- 6. Trapezoidal more accurate for smooth functions

Quiz 07 v6

- 1. $\sin\left(\frac{7\pi}{6}\right) = -\frac{1}{2}$
- 2. $\cos(2\theta) = 2\cos^2\theta - 1$
- 3. $\tan\left(\frac{5\pi}{4}\right) = 1$
- 4. $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$
- 5. Right sum = 104
- 6. Left sum underestimates; right sum overestimates (increasing function)

Quiz 07 – Holistic Grading Rubric

Essential Skills (Front Page – 75%)

Grade	Description
A	Shows correct trigonometric structure and reasoning. All six trig functions correctly evaluated when required. Identities and equations solved with precision. Minor arithmetic errors at most.
B	Correct approach and structure shown. Some computational or sign errors. Minor gaps in algebraic justification.
C	Partial understanding shown. Correct idea attempted but identity use, quadrant reasoning, or solution sets incomplete.
D	Minimal correct reasoning. Major structural misunderstandings in identities or equation solving.
Not Yet	No meaningful evidence of understanding.

Above & Beyond (Back Page – 25%)

Grade	Description
A	Correct setup and execution of Riemann or trapezoidal sums. Accurate use of Δx and function values. Clear mathematical justification when required.
B	Correct structure and setup shown. Minor arithmetic errors.
C	Correct idea attempted but incomplete setup or misapplied formula.
D	Limited understanding of approximation method.
Not Yet	No meaningful attempt.

Teacher Notes

- Must show work for full credit.
- Precision with signs and quadrant reasoning is essential.
- All solutions must include complete solution sets when required.
- Front page evaluates core trig objectives.
- Back page evaluates modeling and approximation reasoning.